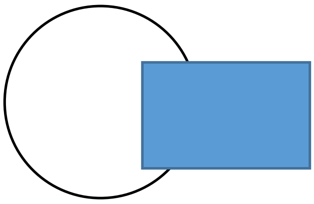
A thin, metallic spherical shell contains a charge Q on it. A point charge q is placed at the center of the shell and another charge q_1 is placed outside it as shown in figure. All the three charges are positive. The force on the charge at the center is

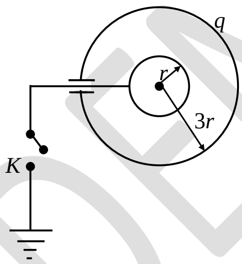
$$(x+a)^n = \sum_{k=0}^{n} \binom{n}{k} x^k a^{n-k}$$



(a) towards left (b) towards right

(c) upward (d) zero

7. Figure shows two conducting thin concentric shells of radii r and $3r$. The outer shell carries charge q, inner shell is neutral. The amount of charge which flows from inner shell to the Earth after the key K is closed is equal to



(a) $-\frac{q}{3}$ (b) $\frac{q}{3}$

(c) $3q$ (d) $-3q$

(a) $\phi_1 = 0$ (b) $0 < \phi_1 < \phi$

(c) $\phi_1 = \phi$ (d) $\phi_1 > \phi$